Waves in Thermoelastic and Multiphysics Continua

Scope brief: the theories and the materials the review covers

164 pages · 38 numbered sections · 90 numbered equations · 25 tables · 40 figures · 337 references · compiled 2 November 2026

This brief says what the review contains, in two parts: which heat-conduction and continuum theories it treats, and which materials it puts numbers on. Every figure printed here is copied from the paper's own tables — the parameter table of §3, the materials table of §3, the comparative tables of §33, the anisotropy table of §14 and the regime table of §30 — so the brief adds no claim, no reference and no number of its own. Where the paper labels a value as a rounded handbook number for illustration, this brief says so too.

1. The organising idea: one template, not a list of theories

The literature presents generalized thermoelasticity one theory at a time, and the labels hide what actually differs between two papers. The review starts instead from a single heat-flux function with two conductivities and three lags:

$$G(s) = [\, k\,(1 + \tau_T\, s) + k^*\,(1 + \tau_v\, s)/s \,] \,/\, (1 + \tau_q\, s),\quad s = -i\omega$$

Set the parameters and the familiar theories appear: Fourier (Biot) at $k^* = \tau_q = \tau_T = \tau_v = 0$, Cattaneo/Lord–Shulman at $\tau_q = \tau_0$, dual-phase-lag with τ_q and τ_T, three-phase-lag when k^* is kept, the Moore–Gibson–Thompson flux form with its third order in time, and the Green–Naghdi types as the limits $k \to 0$ (type II) and lags $\to 0$ with $k^* \neq 0$ (type III). Two theories are deliberately left outside the template, because they change structure rather than a parameter: Green–Lindsay, which modifies the coupling with two relaxation times, and Guyer–Krumhansl, which comes from phonon kinetics and carries a length scale.

Coupling the template to momentum balance gives one dispersion polynomial in $K = k^2$, checked symbolically. From that single polynomial the review reads phase speed, attenuation and thermoelastic damping for every member of the family, and then carries the same machinery into transient, surface, interface, size-effect and multiphysics problems. The exact delay, fractional powers and memory kernels are treated where they make the dispersion relation transcendental or multivalued; every quantitative figure in the paper is produced by an accompanying script.

2. Heat-conduction theories covered

The first block are the members of the template; the second block are the models that leave it and are handled on their own terms. The last column follows the review's own criterion for a rational heat law: $sG(s)$ must tend to a finite non-zero limit as $s \to \infty$.

Theory	Position in the template	Free parameters	Finite thermal front
Classical (Fourier, Biot)	$k^* = 0$, all lags zero	—	no
Lord–Shulman (Cattaneo)	$\tau_q = \tau_0$, $k^* = 0$	τ_0	yes, $c_T = \sqrt{(k/\rho c \tau_0)}$
Dual-phase-lag, first order in τ_q	τ_q, τ_T kept	τ_q, τ_T	no (strictly diffusive)
Dual-phase-lag, second order in τ_q	same law, expanded to $O(\tau_q^2)$	τ_q, τ_T	depends; stable only if $\tau_T \geq \tau_q/2$
Three-phase-lag	k, k^*, τ_q, τ_T, τ_v	four	no (strictly diffusive)
Moore–Gibson–Thompson (flux form)	k, k^*, τ_q	2–3	yes: c_* low, c_T high frequency
Green–Naghdi II	$k = 0$, $k^* > 0$	k^*	yes: $c_* = \sqrt{(k^*/\rho c)}$, non-dispersive
Green–Naghdi III	k, k^* kept, no lags	k, k^*	no (strictly diffusive)
Green–Lindsay	outside: modifies the coupling	τ_0, τ_1	yes
Exact delay (untruncated)	outside: $G = k{\cdot}e^{\wedge}(-\tau_q\, s)$	τ_q	yes, exact front (G not rational)
Caputo-fractional	outside: order α in time	α and a lag	depends on α
Memory-dependent derivative	outside: kernel $K(t - s)$	1–3	variant-dependent
Guyer–Krumhansl (kinetic)	outside: phonon hydrodynamics	l (mean free path), τ_R, τ_N	yes (second sound)

Table 1. Heat-conduction theories and their place in the review's template (paper §3, the parameter table; §33, the comparative table). Thresholds are the paper's: for a rational G, a finite front needs sG(s) → const ≠ 0 as s → ∞.

Three results in this part carry the weight. First, the widely repeated dual-phase-lag stability condition $\tau_T \geq \tau_q/2$ belongs to the second-order Taylor truncation, not to the first-order template, which is stable for every $\tau_T > 0$; the review locates the condition and shows what it does and does not protect. Second, the reductions are checked symbolically and the review records which are regular and which are singular: dual-phase-lag to Lord–Shulman needs $\tau_T \to 0$ at fixed τ_q, not $\tau_q \to 0$, and fractional order $\alpha \to 2$ is the wave equation, not Fourier's law. Third, Taylor truncation of the delay is a modelling choice with a measurable error, so the truncated laws are compared against the exact delay $G = k \cdot e^{(-\tau_q s)}$, with relative error $\approx (\omega\tau)^2/4$ at small $\omega\tau$.

2.1 Elastic and continuum enrichments (size effects)

- Eringen nonlocal elasticity in both forms used in the literature — the stress smoothed by the nonlocal operator and the thermal stress left local, and the form where the coupling is smoothed too; the review shows that the two give different damping near the elastic cut-off.
- Strain-gradient elasticity, with the opposite sign of dispersion to nonlocal elasticity: a positive gradient coefficient stiffens the short waves, whereas the lattice softens.
- Couple-stress and micropolar continua, Klein–Gordon type enrichment, and Kelvin–Voigt viscoelasticity, each used in the recent thermoelastic literature.
- Nonlocality in the heat flux as well as in the stress, and the caveat that the nonlocal length ξ must be calibrated against a stated target (zone boundary or long wavelength), since the two calibrations do not agree.

2.2 Coupled fields

Four added fields are developed to the point where they can be solved, not merely listed:

- Piezoelectricity and thermopiezoelectricity: bulk stiffening, the exact renormalisation of the thermoelastic dispersion relation by the electric field, pyroelectric sign conventions, and Bleustein–Gulyaev shear-horizontal surface waves on a hexagonal (6mm) crystal with metallised and free electrical boundary conditions.
- Magneto-thermoelasticity: the magnetic field enters as a stiffness contribution at all frequencies.
- Thermoelastic diffusion and chemical coupling.
- Microstructure — micropolar, voids and porosity — and the two-temperature and semiconductor/photothermal variants treated in the taxonomy.

2.3 Wave problems worked out

Problem	What the review reports
Bulk plane waves in the template family	dispersion polynomial roots, phase speed and attenuation for each member, branches tracked by continuity
Anisotropy (cubic crystals)	coupling of quasi-longitudinal waves along [100] and [111]; for copper the coupling itself differs by 41 % between the two directions
Reflection at a thermoelastic free surface	amplitude and energy coefficients, elastic limit recovered to 10^{-10}
Thermoelastic Rayleigh waves	speed shift $\approx 2\times10^{-3}$ at low wavenumber for $\varepsilon = 0.05$, $\nu = 0.3$; damping peak $Q^{-1} \approx 2\times10^{-3}$ at $k\, c_1/\omega_* \approx 0.4$; Lord–Shulman changes the damping by less than 1 % at low k and up to ≈ 12 % at high k
Thermoelastic Lamb waves in a plate	6×6 determinant; elastic limit and the Lifshitz–Roukes flexural loss recovered with ratio 1.000 for small kh
Clamped-bar modes	modal loss of a bar against the exact Fourier solution
Interfaces and energy partition	when the mode-wise energy ratios sum to one and when they do not; net heat-flux continuity as the correct bookkeeping
Layered media	Love, Stoneley and Scholte-type waves; classical closed forms used as benchmarks
Transients	heat pulse in a half-space (Fourier versus Cattaneo, exact front) and thermal shock of a coupled half-space, with the boundary conditions satisfied to machine precision
Thermal-wave resonance	resonances of a finite rod and the Cattaneo mode transition at $k_c = 1/(2\sqrt{(\chi\tau)})$
Flexural thermoelastic damping	Zener relaxation and the exact Lifshitz–Roukes result, with the plane-strain factor $(1 + \nu)/(1 - \nu)$

Table 2. The wave and damping problems solved in the review (sections 13–26). The quoted values are the ones printed in those sections; all are computed from rounded handbook inputs.

3. Materials covered

Only five materials carry computed numbers, and the paper labels those numbers as handbook-type values at 300 K, to be replaced by data for the specimen at hand:

Material	c_1 (m/s)	ε (coupling)	$f_* = \omega_*/2\pi$ (Hz)	$\alpha = k/\rho c$ (m²/s)
Aluminium	6198	0.027	6.3×10^{10}	9.8×10^{-5}
Copper	4540	0.019	2.8×10^{10}	1.2×10^{-4}
Silicon	8854	0.00056	1.4×10^{11}	9.2×10^{-5}
Steel (carbon)	5778	0.0097	4.1×10^{11}	1.3×10^{-5}
Fused silica	5972	7.2×10^{-6}	6.6×10^{12}	8.6×10^{-7}

Table 3. Thermoelastic coupling and characteristic frequency for the five materials used in the calculations (paper §3, materials table; rounded handbook-type values at 300 K).

Copper, aluminium and silicon are then treated as cubic crystals, where the coupling is direction-dependent:

Crystal	ε along [100]	speed shift [100]	ε along [111]	speed shift [111]	ratio
Cu	0.0238	1.18 %	0.0169	0.84 %	1.41
Al	0.0325	1.61 %	0.0306	1.52 %	1.06
Si	6.5×10^{-4}	0.032 %	5.3×10^{-4}	0.026 %	1.23

Table 4. Direction dependence of thermoelastic coupling for quasi-longitudinal waves (paper §14; 300 K, rounded handbook inputs).

3.1 Materials discussed without computed numbers

The evidence and applications sections name further materials, deliberately without new numbers, because the point there is what the experiment shows rather than a prediction:

- Solid helium and sodium fluoride — the classical second-sound observations, in narrow temperature windows where momentum-conserving phonon collisions dominate.
- Graphite — second sound above 100 K, the one case where a hydrodynamic phonon window is claimed at high temperature; the review quotes the source's fitted speed at 85 K (≈ 3200 m/s against 3650 m/s from ab initio) and records that no error bar is given.
- Hexagonal 6mm piezoelectric crystals — the Bleustein–Gulyaev surface wave, with electromechanical coupling $K^2 = e_{15}^2/(c_{44}\varepsilon)$.
- Layered and bonded pairs — substrate–film, electrode–crystal, welded metal–metal, treated through impedance ratios rather than named materials.
- Saturated and partially saturated porous solids, and biological tissue — the latter described through the Pennes bioheat baseline and the disputed reports of relaxation times of order seconds.
- Water-like media, used only to show that $\omega_* \approx 10^{13}$ rad/s lies far above any frequency of interest for tissue.

4. Which theory for which material and frequency

This is the practical answer the review gives, and it is decided by two dimensionless groups: $\omega\tau_q$, which measures how far the frequency is from the flux lag, and ω/ω_*, which measures how far it is from the heat-carrier relaxation. Six scenarios are evaluated with $\tau_q = 1$ ps:

Scenario	Material	f (Hz)	ω/ω_*	$\omega\tau_q$	δ_T (m)	λ (m)	Assessment
Ultrasonic testing	Aluminium	1.0×10^{7}	1.6×10^{-4}	6.3×10^{-5}	1.8×10^{-6}	6.2×10^{-4}	Fourier adequate
GHz acoustics	Aluminium	1.0×10^{9}	1.6×10^{-2}	6.3×10^{-3}	1.8×10^{-7}	6.2×10^{-6}	Fourier adequate
Picosecond ultrasonics	Aluminium	1.0×10^{11}	1.6	6.3×10^{-1}	1.8×10^{-8}	6.2×10^{-8}	lag may matter; check kinetics
Silicon beam, h = 1 µm, at f_Z	Silicon	1.4×10^{8}	1.1×10^{-3}	9.1×10^{-4}	4.5×10^{-7}	6.1×10^{-5}	Fourier adequate
Silicon film, h = 20 nm, at f_Z	Silicon	3.6×10^{11}	2.7	2.3	9.0×10^{-9}	2.5×10^{-8}	continuum law questionable
Copper film, h = 50 nm, at f_Z	Copper	7.3×10^{10}	2.6	4.6×10^{-1}	2.3×10^{-8}	6.2×10^{-8}	lag may matter; check kinetics

Table 5. Dimensionless groups for six representative scenarios at $T_0 = 300$ K and $\tau_q = 1$ ps (paper §30, regime table; the table is generated by the paper's own script from the rounded handbook inputs). δ_T is the thermal penetration depth, λ the acoustic wavelength.

4.1 The decision rule as stated in the paper

- Fourier is adequate when $\omega\tau_q < 0.1$ and $\omega/\omega_* < 0.1$; in that corner the classical coupled theory is the baseline for essentially every ultrasonic, above-micron device and geophysical application.
- A lag law may matter when $\omega\tau_q < 1.5$, but then the kinetic validity of the continuum model has to be checked at the same time, because the same corner of the map is where carrier mean free paths rival the wavelength.
- Beyond that, the continuum heat law itself is questionable: at 20–50 nm films the first deviation from Fourier is size-related, not lag-related, and the Guyer–Krumhansl or nonlocal description is the appropriate next model because it carries a length scale.
- Electromechanical and magnetic couplings change stiffness at all frequencies, so they matter even where the heat law does not.

5. Map of the paper

Part	Sections	Content
Foundations	§1–§5	introduction; historical development; balance laws, thermodynamics, the template and its admissibility tests; kinetic origin of the hyperbolic laws
Heat laws	§6–§12	classical coupled theory; Lord–Shulman, Green–Lindsay, Green–Naghdi; dual-, three-phase-lag and MGT; fidelity of the Taylor truncations; fractional and memory-dependent models; causality and Kramers–Kronig relations
Waves	§13–§18	three-dimensional propagation; cubic anisotropy; free-surface reflection; transients and thermal shock; thermal-wave resonance; Rayleigh waves
Structure and size	§19–§26	Lamb waves and bar modes; flexural damping (Zener and Lifshitz–Roukes); interfaces and energy partition; layered Love/Stoneley/Scholte waves; nonlocal and gradient size effects; piezoelectric, magnetic, diffusive and microstructural couplings; Bleustein–Gulyaev waves
Practice	§27–§32	solution methods and verification; a finite-difference benchmark for hyperbolic heat conduction; applications from laser ultrasonics to resonators and tissue; regime table; worked examples; what the experiments actually identify
Synthesis and apparatus	§33–§38, Appendices A–H	comparative synthesis, decision guide, pitfalls checklist; taxonomy of the field; atlas of problems from other groups; reporting standards; open problems; balance-law and dispersion derivations; glossaries; per-equation and benchmark registers; the classified reference catalogue

Table 6. Structure of the review, using the paper's own section numbers.

6. What the brief and the paper do not claim

- No new experiment is reported. Every result is analytical or numerical, and the review says so where the evidence is thin.
- The material numbers are rounded handbook-type inputs for illustration, at 300 K, not measurements of a named specimen.
- Code verification is not physical validation: the benchmarks test that an implementation reproduces an exact or independently computed reference to a stated tolerance, and the paper states explicitly what a pass does not establish.
- Of the works cited, 103 are recorded as SOURCE NOT AVAILABLE in the paper's claim register: nothing is described from them, and no claim rests on them.
- The atlas of problems is a first instalment limited to papers read in full text, not a representative sample of the field.

Provenance of the numbers in this brief

Content here	Taken from
Template, parameter assignments, finite-front criterion	§3, §33 (comparative table), §8
Materials table, coupling and characteristic frequency	§3
Cubic anisotropy of coupling	§14
Six scenarios and their assessment	§30, §33.5 (decision guide)
Rayleigh and Lamb numbers quoted in Table 2	§18, §19
Page, section, equation, table, figure and reference counts	the paper's build and gate log

Every item is a file in the review package; none is from memory.